

Expt. No. 08	Calculation of Secondary turns and Current in a transformer

AIM:

To calculate the secondary turns and current in a single phase transformer.

APPARATUS REQUIRED:

Sl No.	Apparatus	Range	Quantity
1.	Single phase transformer	230/115 V	1
2.	Auto transformer	0-300V AC	1
3.	Voltmeter	0-300V AC	2
	Ammeter	0-10A AC	1
4.	Connecting wires	-	As required
5.	Single phase AC power supply	-	-

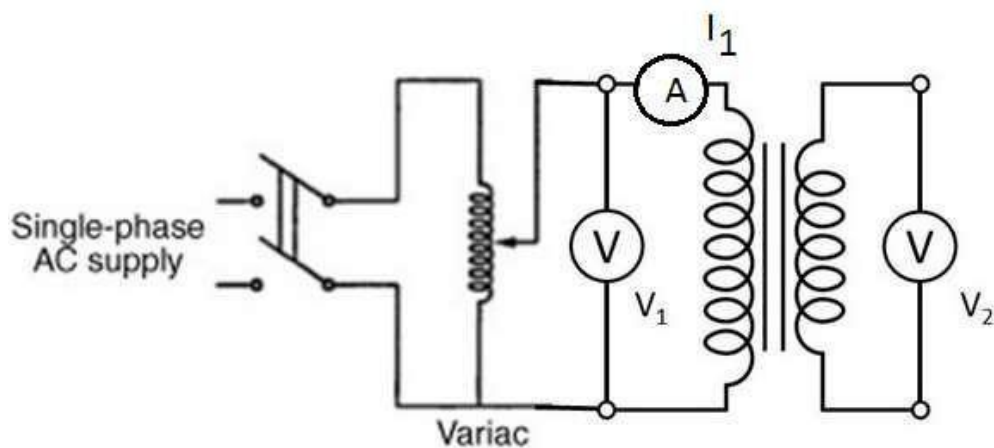
THEORY:

It is essential to know the relative polarity at any instant of primary and secondary terminals for making correct connections. When the two transformers are to be connected in parallel to share the load on the system. The marking is correct if voltage V_3 is less than V_1 , such a polarity is termed as subtractive polarity. The standard practice is to have subtractive polarity because it reduces the voltage stress between adjacent loads. In case $V_3 > V_1$, the EMF induced in primary and secondary have additive relation and transformer is said to have additive polarity

PROCEDURE:

1. Connect the circuit as shown in the diagram.
2. Switch on the a.c. supply.
3. Record voltage V_1 across primary and V_2 across various tapings of secondary.
4. If $V_1 > V_2$ then transformer is step down.
5. If $V_2 > V_1$ then transformer is step up.
6. Switch off a.c. supply

CIRCUIT DIAGRAM:



TABULAR COLUMN:

S.NO	Primary Voltage (V1) in volts	Secondary voltage (V2) in volts	Primary Current (I1) in amps	Transformation ratio ($k=V_2/V_1$)	Secondary current, $I_2=I_1/k$
1	60	30	17	0.5	34
2	80	40	13	0.5	26
3	100	50	10	0.5	20
4	120	60	8	0.5	16

Marks Obtained:

Theoretical Calculations	20	
Observation	20	
Execution of practice examples	30	
Viva	10	
Record	20	
Total Score	100	
Date of experiment		Faculty signature
Date of record submission		

RESULT:

Thus the transformation ratio and the secondary current are measured for a given single phase transformer.

Expt. No. 9	Load Test on a single phase transformer

AIM:

To conduct load test on single phase transformer and to find efficiency and percentage regulation.

APPARATUS REQUIRED:

S.No.	Apparatus	Range	Type	Quantity
1	Ammeter	(0-10)A	MI	1
		(0-5) A	MI	1
2	Voltmeter	(0-150)V	MI	1
		(0-300) V	MI	1
3	Wattmeter	(300V, 5A)	Upf	1
		(150V, 5A)	Upf	1
4	Auto Transformer	1 ϕ , (0-260)V	-	1
5	Resistive Load	5KW, 230V	-	1
6	Connecting Wires	2.5sq.mm	Copper	Few

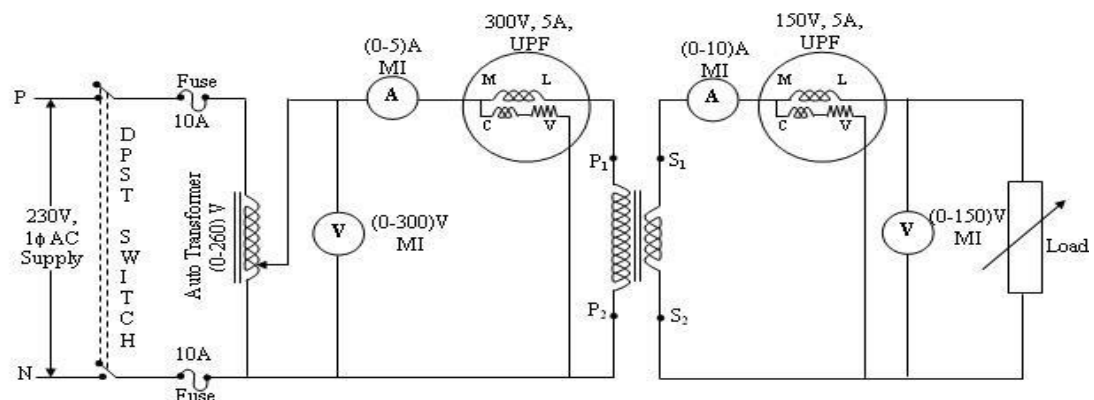
PRECAUTIONS:

1. Auto Transformer should be in minimum position.
2. The AC supply is given and removed from the transformer under no load condition.

PROCEDURE:

1. Connections are made as per the circuit diagram.
2. After checking the no load condition, minimum position of auto transformer and DPST switch is closed.
3. Ammeter, Voltmeter and Wattmeter readings on both primary side and secondary side are noted.
4. The load is increased and for each load, Voltmeter, Ammeter and Wattmeter readings on both primary and secondary sides are noted.
5. Again no load condition is obtained and DPST switch is opened.

CIRCUIT DIAGRAM:



FUSE RATING:

125% of rated current

$$\frac{125 \times 5}{100} = 6.25 \text{ A}$$

NAME PLATE DETAILS:

	Primary	Secondary
Rated Voltage :	230V	115V
Rated Current :	5A	10 A
Rated Power :	1KVA	1KVA

TABULAR COLUMN:

S.No	Load Current (I ₂)	Secondary Voltage (V ₂)	Primary Voltage (V ₁)	Primary Current (I ₁)	Power Input	Power Output	Efficiency (P _o /P _i) %	Voltage Regulation (%)
1.	0	230	230	0	0	0	0	$\frac{230-230}{230} = 0$
2.	0.3	220	228	1	212	200	94.3	$\frac{230-220}{220} = 4.5\%$
3.	0.8	216	228	1.8	500	480	96	$\frac{230-216}{216} = 6.4\%$
4.	2.1	212	224	2.6	740	720	97.2	$\frac{230-212}{212} = 8.4\%$
5.	3	208	224	3.4	960	940	98.9	$\frac{230-208}{208} = 10\%$

FORMULAE:

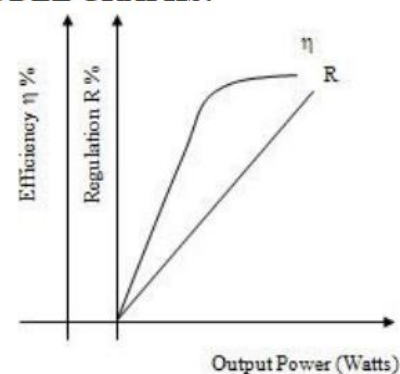
Output Power = W₂ x Multiplication factor

Input Power = W₁ x Multiplication factor

$$\text{Efficiency } \eta \% = \frac{\text{Output Power}}{\text{Input Power}} \times 100\%$$

$$\text{Regulation } R \% = \frac{V_{NL} - V_{FL} (\text{Secondary})}{V_{NL}} \times 100\%$$

MODEL GRAPHS:



Marks Obtained:

Theoretical Calculations	20	
Observation	20	
Execution of practice examples	30	
Viva	10	
Record	20	
Total Score	100	
Date of experiment		Faculty signature
Date of record submission		

RESULT:

Thus the load test on single phase transformer is conducted.

Expt. No. 10	OC and SC test on single phase transformer

AIM:

To conduct OC and SC test on single phase transformer.

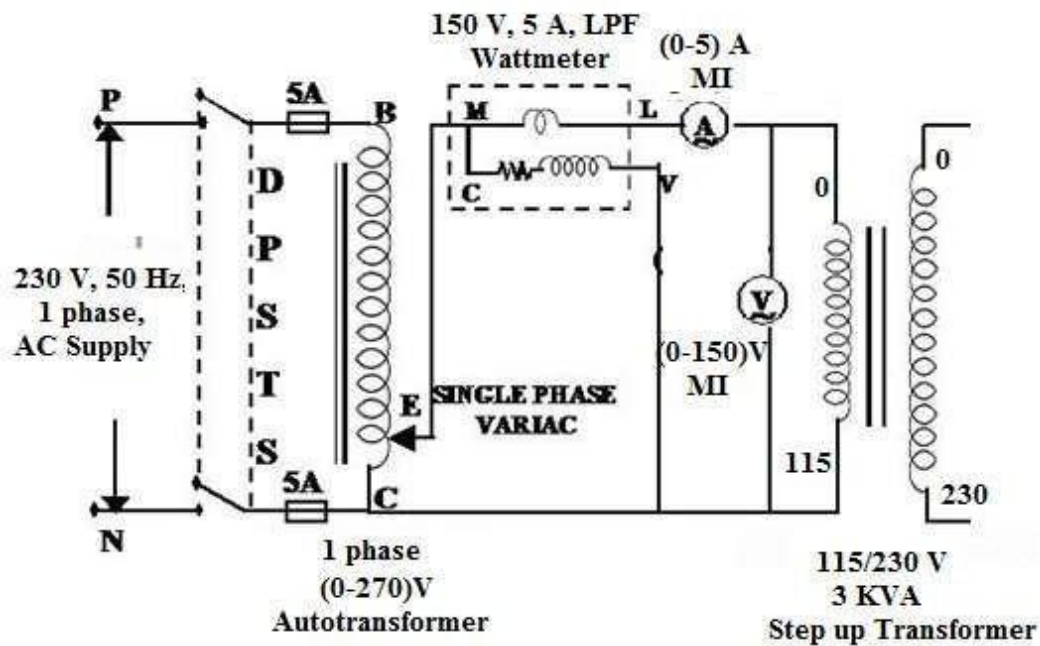
APPARATUS REQUIRED:

S.No.	Apparatus	Range	Type	Quantity
1	Ammeter	(0-10)A	MI	1
		(0-5) A	MI	1
2	Voltmeter	(0-150)V	MI	1
		(0-300) V	MI	1
3	Wattmeter	(300V, 5A)	Upf	1
		(150V, 5A)	Upf	1
4	Auto Transformer	1 ϕ , (0-260)V	-	1
5	Resistive Load	5KW, 230V	-	1
6	Connecting Wires	2.5sq.mm	Copper	Few

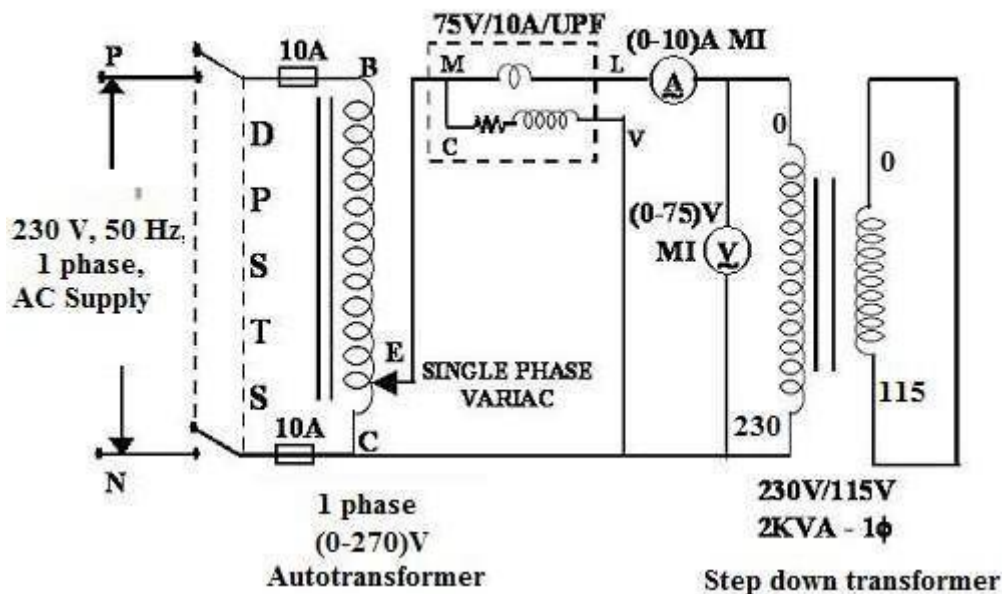
PRECAUTIONS:

1. Auto Transformer should be in minimum position.
2. The AC supply is given and removed from the transformer under no load condition.

OPEN CIRCUIT TEST



SHORT CIRCUIT TEST



PROCEDURE:

I) O.C. TEST:

1. The circuit connections are made as per the circuit diagram.
2. Keeping the H.V. winding open and the autotransformer in its minimum position the main supply is switched ON.
3. By slowly and carefully adjusting the autotransformer, the rated voltage (115V) is applied to L.V. winding of the transformer.
4. Under this condition the ammeter (I_o), Voltmeter (V_o) and Wattmeter (W_o) readings are noted down.
5. After the experiment is completed, the autotransformer is slowly brought back to minimum position and then the main supply is switched OFF.

II) S.C. TEST:

1. The circuit connections are made as per the circuit diagram.
2. Short circuiting the LV winding and keeping the autotransformer in its minimum position, the main supply is switched ON.
3. By slowly and carefully adjusting the autotransformer the rated current (which is calculated as $H.V. \text{ current} = \frac{\text{KVA Rating} \times 1000}{H.V.}$) is circulated through the H.V. winding.
4. Under this condition, the ammeter (I_{SC}), the voltmeter (V_{SC}) and the Wattmeter

(WSC) readings are noted down.

5. After the experiment is completed, the autotransformer is brought back to its minimum position and main supply is switched OFF.

TABULATION:

TEST	VOLTAGE(V)	CURRENT(A)	POWER(W)
O.C TEST (On L.V side)	$V_o = 230V$	$I_o = 0.3A$	$W_o = 64W$
S.C. TEST (On H.V side)	$V_{sc} = 24V$	$I_{sc} = 4.2A$	$W_{sc} = 76W$

Marks Obtained:

Theoretical Calculations	20	
Observation	20	
Execution of practice examples	30	
Viva	10	
Record	20	
Total Score	100	
Date of experiment		Faculty signature
Date of record submission		

RESULT:

Thus the O.C and S.C tests are conducted on the single phase transformer.

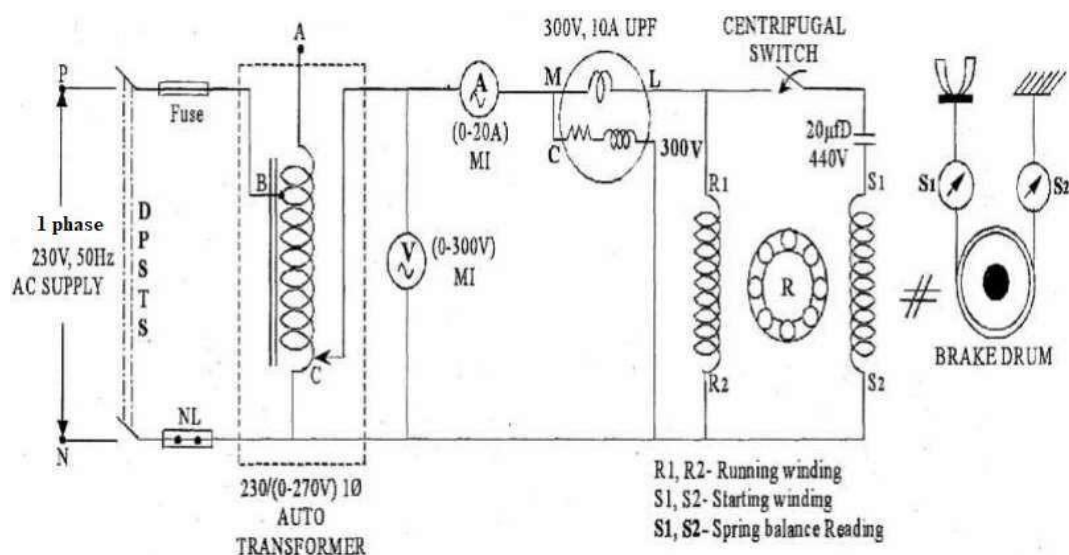
Expt. No. 11**Load Test on a Single Phase Induction Motor****AIM:**

To conduct load test on single phase induction motor to plot the following performance characteristics

- 1) Output power vs Efficiency
- 2) Output power vs speed
- 3) Output power vs % slip
- 4) Output power vs Torque
- 5) Output power vs Line current

APPARATUS REQUIRED:

S.No.	Apparatus	Range	Type	Quantity
1	Ammeter	(0-20)A	MI	1
2	Voltmeter	(0-300) V	MI	1
3	Wattmeter	(300V, 10 A)	Upf	1
4	Auto Transformer	1 ϕ , (0-270)V	-	1
5	Connecting Wires	2.5sq.mm	Copper	Few

CIRCUIT DIAGRAM**PROCEDURE:**

1. The circuit connections are made as per the circuit diagram.
2. Keeping the motor field rheostat in its minimum position and the starter in its OFF

position the main supply is switched ON to the circuit.

3. The motor is started using the three point starter by slowly and carefully moving the starter handle from it's OFF to ON position.
4. The motor is brought to its rated speed by gradually adjusting the field rheostat and checked with the help of a tachometer.
5. Under this no load condition one set of readings namely, applied voltage (V_L), line current (I_L), the two spring balance readings (S_1 and S_2) and motor speed (N) are noted down in the tabular columns.
6. The load on the motor is increased in steps gradually and at each step, all the meter readings and the motor speed are recorded in the tabular column. The above procedure is repeated until the motor is loaded to 120% of its rated current.
7. After the experiment is completed, the load on motor is gradually decreased to minimum and the rheostat is brought back to its original position and then the main supply is switched OFF.
8. Plot the performance characteristics

TABULAR COLUMN:

S.No	Volts (V)	Amps (A)	Speed	Wattmeter Reading	Spring Balance Reading			Torque $= 9.81 \times R \times (S_2 - S_1)$	Input (W)	Output $\frac{2 \pi T N}{60}$	Power Factor $\frac{W}{VI}$	$\eta = \frac{O/P}{I/P}$	$S = \frac{N_s - N}{N_s}$
					S_1	S_2	$S_2 - S_1$						
1	230	3.2	1490	300	0	0	0	0	736	736	0.407	0.4	0.006
2	230	3.5	1480	540	0.8	1	0.2	0.196	805	30.392	0.670	3.77	0.03
3	228	4	1460	760	1.6	2	0.4	0.392	912	59.963	0.833	6.37	0.036
4	226	4.5	1450	900	2	2.8	0.8	0.784	1017	119.106	0.884	11.71	0.033

Model Calculations:

$T = 9.81 \times 0.1 \times 0.2$
 $T = 0.1962 \text{ Nm}$

Input Power = 230×3.5
 $= 805 \text{ W}$

Output Power = $\frac{2 \times 3.14 \times 0.1962 \times 1480}{60}$
 $\text{Power} = 30.392 \text{ W}$

PF = $\frac{540}{230 \times 3.5}$
 $PF = 0.670$

$\eta = \frac{O}{I} = \frac{30.392}{805}$
 $\eta = 3.77\%$

Slip = $\frac{1500 - 1480}{1500}$
 $\text{Slip} = 0.013$

MODEL CALCULATIONS:

The **circumference of the brake drum** is measured and the radius of the drum is calculated using the expression

$$r = C/2\pi = \underline{\hspace{2cm}} \text{ (m)}$$

r - Radius of the Brake drum (m), c – Circumference of the Brake Drum (m)

The **torque** developed by the motor is given by

$$T = (S_1 - S_2) * R * 9.81 = \underline{\hspace{2cm}} \text{ (Nm)}.$$

The **output power** of the motor is

$$P_{out} = 2\pi NT/60 = \underline{\hspace{2cm}} \text{ (W)}.$$

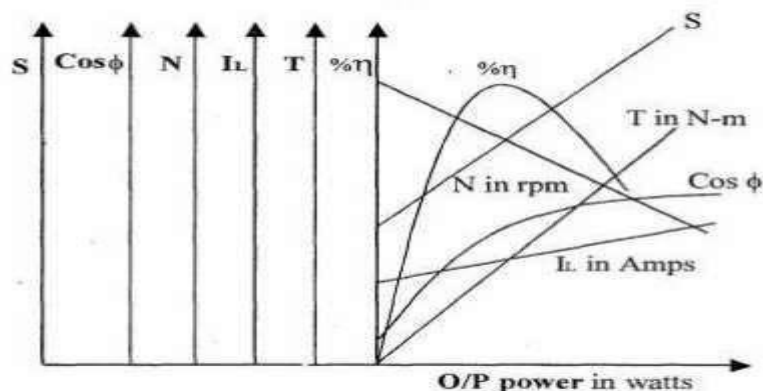
The **efficiency** of the motor

$$\eta = (P_{out} / P_{in}) * 100 = \underline{\hspace{2cm}}$$

The **%slip** of the motor

$$\% s = (N_s - N / N_s) * 100 \text{ where } N_s \text{ --- Synchronous speed} = (120 * f) / P = \underline{\hspace{2cm}} \text{ (rpm)}.$$

MODEL GRAPH:



Marks Obtained:

Theoretical Calculations	20	
Observation	20	
Execution of practice examples	30	
Viva	10	
Record	20	
Total Score	100	
Date of experiment		Faculty signature
Date of record submission		

RESULT:

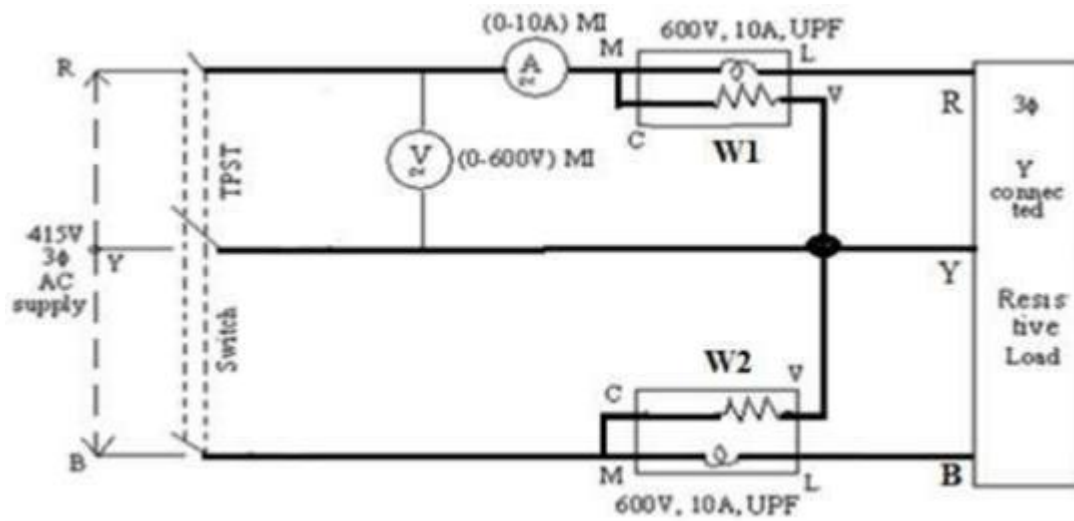
Thus the load test on single phase Induction motor is conducted and their performance characteristics are plotted.

AIM:

To measure the power in three phase balanced star connected load using two wattmeter method.

APPARATUS REQUIRED:

S.No.	Apparatus	Range	Type	Quantity
1	Ammeter	(0-10)A	MI	1
2	Voltmeter	(0-600) V	MI	1
3	Wattmeter	(600V, 10A)	Lpf	2
4	Auto Transformer	3 ϕ , (0-600)V	-	1
5	Lamp load	-	-	1
6	Connecting Wires	2.5sq.mm	Copper	Few

CIRCUIT DIAGRAM**PROCEDURE:**

1. Make the connections as per the circuit diagram shown in Fig 5.1.
2. Switch ON 3 phase AC supply.
3. Apply load and measure the values of wattmeters, ammeter and voltmeter.
4. Switch OFF all the loads and supply.

TABULAR COLUMN:

Sl No.	Line voltage V_L (V)	Line current I_L (A)	Wattmeter (W1)	Wattmeter (W2)	$W = W1 + W2$	Calculated power (W)
1	175	2	320	280	600	606.2
2	185	2.1	335	347	682	672.9
3	195	2.2	370	390	760	753.2
4	205	2.3	410	430	840	830.9

MODEL CALCULATIONS:

1. Total Power, $P = W1 + W2$ Watts.
2. Power Factor $\cos \phi = \cos [\tan^{-1} \{ \sqrt{3} (W1 - W2) / (W1 + W2) \}]$
3. Calculated Power $= \sqrt{3} V_L * I_L * \cos \phi$

Marks Obtained:

Theoretical Calculations	20	
Observation	20	
Execution of practice examples	30	
Viva	10	
Record	20	
Total Score	100	
Date of experiment		Faculty signature
Date of record submission		

RESULT:

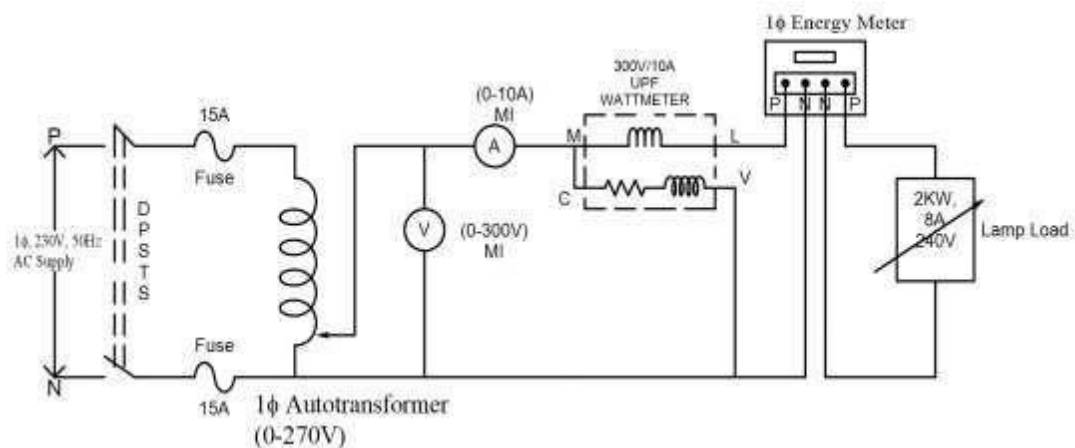
Thus, the power in three phase balanced star connected load using two wattmeter method is measured and calculated.

Expt. No. 13**Energy consumption measurement using a Single Phase Energy Meter****AIM:**

To measure the energy consumption using single phase energy meter.

APPARATUS REQUIRED:

S.No.	Apparatus	Range	Type	Quantity
1	Ammeter	(0-20)A	MI	1
2	Voltmeter	(0-300) V	MI	1
3	Auto Transformer	1 ϕ , (0-270)V	-	1
4	Lamp load	-	-	1
5	Connecting Wires	2.5sq.mm	Copper	Few

CIRCUIT DIAGRAM**PROCEDURE:**

1. Connect the circuit as shown in circuit diagram.
2. Connect the load rheostat 10 Ω /12A in series with load.
3. Keep the variac in minimum output voltage position.
4. Keep the load in maximum position.
5. Adjust the variac output equal to the rated voltage of energy meter.
6. Adjust the load till rated current of energy meter passes through it.
7. Note down the voltmeter and ammeter readings.
8. Note down the time taken for 2 revolutions of disc in the energy meter.
9. Switch off the supply.
10. Repeat the above steps for different currents.

TABULAR COLUMN:

S.No	Line Voltage V_L (V)	Line current I_L (A)	Time taken for 10 revolutions (sec)	No. of revolution (n)	Energy Calculated (kwhr)	Measured Energy (kwhr)
1	230	1	85 sec	10	5.43×10^{-3}	6.66×10^{-3}
2	225	3	38 sec	10	7.125×10^{-3}	6.66×10^{-3}

Model Calculations:

Calculated Energy = $\frac{V \times I \times t}{3600} = \frac{5.43}{1000} = 5.43 \times 10^{-3} \text{ kWh}$

Measured Energy = $\frac{n}{1500 \text{ rev/kwh}} = \frac{10 \text{ rev}}{1500 \text{ rev/kwh}} = 6.66 \times 10^{-3} \text{ kWh}$

MODEL CALCULATIONS:

- Calculated energy consumed during 'n' revolutions $W_a = V \cdot I \cdot t / 3600 =$ (watt-hour)
 where V= Voltage (V)
 I= Current (A)
 t = Time (S)
- Measured Energy by energy meter $W_R = n / \text{meter constant}$

Marks Obtained:

Theoretical Calculations	20	
Observation	20	
Execution of practice examples	30	
Viva	10	
Record	20	
Total Score	100	
Date of experiment		Faculty signature
Date of record submission		

RESULT:

Thus the energy consumption using single phase energy meter is measured.

Expt No 14

Load Test on DC shunt Motor

AIM:

- To conduct load test on DC shunt motor and to determine the efficiency using mechanical loading arrangement
- To Draw the following characteristics curves
- Torque Vs Armature current (electrical characteristics)
- Speed Vs Armature current
- Speed Vs Torque (Mechanical characteristics)
- Performance curve

Apparatus Required:

Sl. No.	Name	Specification	Quantity
1	DC shunt motor with spring balance load	2 HP, 1440 rpm, 230V, 18.1A	1 no.
2	Voltmeter	(0-300) V, PMMC	1 no.
3	Ammeter	(0-20) A, PMMC	1 no.
4	Rheostat	1250 Ω , 0.8A	1 nos.
5	Tachometer	Digital Type	1 no.
6	Connecting wires	PVC Insulated copper	as per required

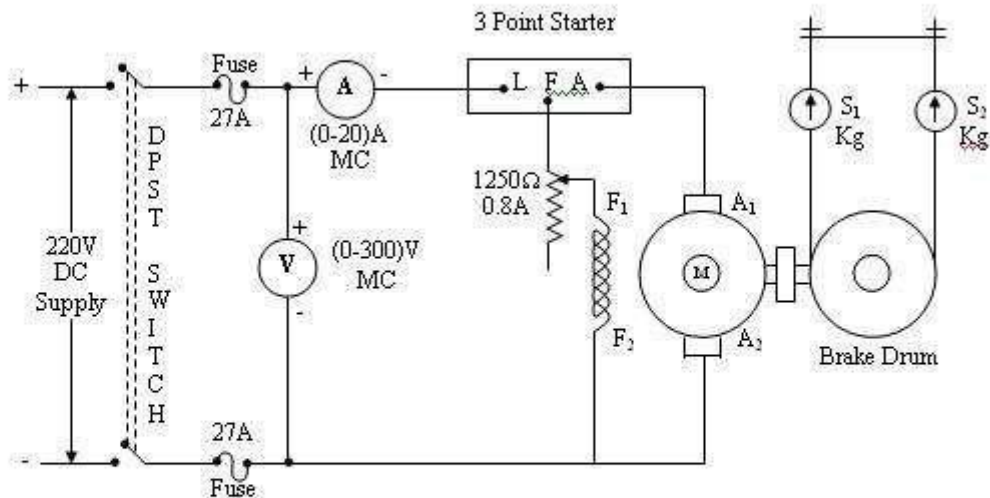
PRECAUTIONS:

- DC shunt motor should be started and stopped under no load condition.
- Field rheostat should be kept in the minimum position.
- Brake drum should be cooled with water when it is under load.

PROCEDURE:

- Connections are made as per the circuit diagram
 - After checking the no load condition and minimum field rheostat position starter resistance is gradually removed.
 - The motor is brought to its rated speed by adjusting the field rheostat.
 - Ammeter, Voltmeter readings, speed and spring balance readings are noted under no load condition.
 - The load is then added to the motor gradually and for each load, voltmeter, ammeter, spring balance readings and speed of the motor are noted (Take readings up to rated current of the machine).
 - The motor is then brought to no load condition and field rheostat to minimum position, then switch off the supply.
-

CIRCUIT DIAGRAM:



FUSE RATING:

125% of rated current

$$\frac{125 \times 21}{100} = 26.25A$$

NAME PLATE DETAILS:

Rated Voltage : 220V
Rated Current : 21A
Rated Power : 3.5KW
Rated Speed : 1500 RPM

FORMULAE USED:

▪ Torque (T) = 9.81 (S₁-S₂). r

Where, S₁, S₂ → Spring balance readings in kg

r → Radius of brake drum in m

9.81 → Constant to convert kg to Newton

2. Output Power (P_{out}) = $\frac{2\pi NT}{60}$

Where N = Speed in rpm

3. Input power (P_{in}) = $V_L \cdot I_L$ Watts

V_L = Input voltage in volts

I_L = Input current in Amps

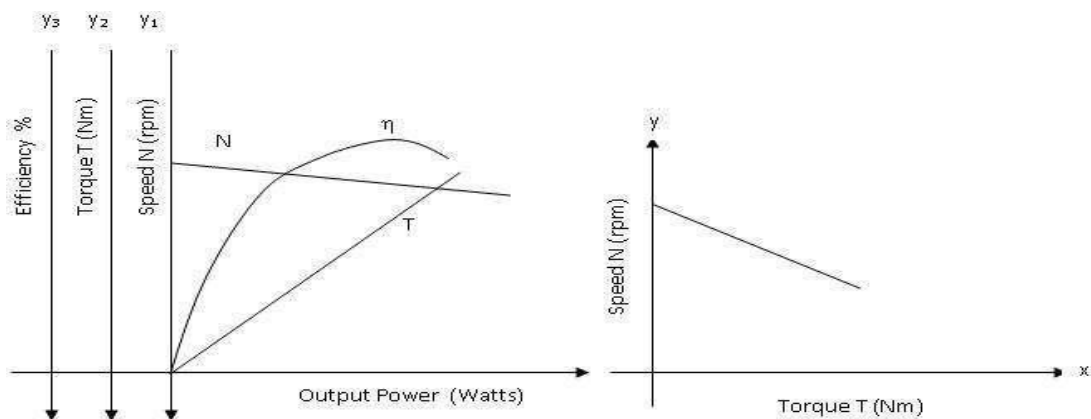
4. % Efficiency = $(P_{out}/P_{in}) \cdot 100$

Observation Table: –

Radius of the Brake drum in meter =

S.No	Voltage (V)	Current (A)	Speed (N)	Spring Balance		$f_1 = S_1 - S_2$ 9.81 $T = F_1 \times R$ (Torque)	I/P power $V_L \times I_L$	O/P power $\frac{2\pi T N}{60}$	Efficiency in % OP/IP
				S_1	S_2				
1	220	1.6	1450	0	0	0	352	0	0%
2	220	4	1422	0.6	2.2	1.412	880	210.15	23.8%
3	220	6	1400	0.8	5.2	3.884	1320	569.135	43.1%
4	220	8	1382	1	8	6.1803	1760	893.9	50.7%
5	218	10	1378	1.2	10.2	7.946	2180	1146.06	52.5%
6	217	12	1359	1.8	13	9.88	2406	1406.4	58.4%

Model Graph:



Marks Obtained:

Theoretical Calculations	20	
Observation	20	
Execution of practice examples	30	
Viva	10	
Record	20	
Total Score	100	
Date of experiment		Faculty signature
Date of record submission		

Result:

Thus the Load Test on DC shunt Motor is experimentally verified.